

Ekansh Khanulia

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EDUCATION

Leiden University

- **Master of Science** | Computer Science (specialization in Data Science)
- Relevant coursework: Information Retrieval, Text & Data Mining, Reinforcement & Deep Learning

Leiden, Netherlands

Feb 2025 - Present

Graphic Era Hill University

- **Bachelor of Technology** | Computer Science (CGPA: 8)
- Relevant coursework: Engineering Mathematics, Data Structures, Design Algorithms and Analysis, DBMS

Dehradun, India

June 2019 - June 2023

EXPERIENCE

Summer Trainee

June 2023 – Aug 2023

Defence Research and Development Organization, DEAL | [Link](#)

Dehradun, India

- Built a real-time facial recognition and drowsiness detection system for monitoring staff alertness at the DRDO facility.
- Engineered an integrated pipeline using OpenCV, dlib, and Haar cascades for face, eye, and gaze tracking; implemented EAR-based alert triggers and trained a local LBPH classifier on collected data for identity recognition.
- Monitored 150+ hours of internal checkpoint activity using a real-time AI pipeline, identifying 35+ attention loss events with 88% accuracy, reducing manual supervision and improving response time.

PROJECTS

Cartpole-Deep RL-Benchmark-Suite | [Link](#)

March 2025 – May 2025

- Built a RL pipeline to benchmark REINFORCE, AC, A2C, and PPO on CartPole-v1, using shared hyperparameters and naïve baseline.
- Applied PPO optimizations (GAE, entropy bonus, ϵ clipping, mini batches), boosting training speed and reward stability by 2.5x over A2C.
- Achieved maximum episodic reward (500) within 200K steps, outperforming baseline PPO by 2x and A2C by 3x; reduced training variance by 50%, improving policy robustness and generalization.

Flappy-RL-Benchmarks | [Link](#)

May 2025 – June 2025

- Built a Pygame Flappy Bird environment with varying difficulty to test RL agents.
- Implemented DQN, DDQN, SAC, and PPO in the environment using prioritized replay, GAE, entropy regularization, and naive baselines, with tuned rewards and adjusted pipe speed to simulate difficulty levels (−6 to −13 pixels/frame).
- PPO outperformed all agents, achieving a max score of 1200, highest average reward (892), and converging 2 times faster than SAC, demonstrating stable and efficient learning across difficulty settings.

Pacman-rangers | [Link](#)

March 2025 – April 2025

- Built an advanced MCTS agent with heuristic rollouts and biasing multi-agent Pacman Capture the Flag.
- Implemented ablation study on hyperparameters (rollout depth, C_p , ϵ), heuristic state evaluations, untried action sorting, and weighted simulations; integrated findings into a “Rainbow” MCTS for optimized end-to-end performance.
- Achieved a TrueSkill score of 35.6 with the best-performing agent, outperforming baseline MCTS (score: 7.1) and heuristic agents (score: 25.9) in a 700-match tournament, with a 65%+ win rate, demonstrating robust generalization.

AWARDS & ACHIEVEMENTS

- Secured a coveted Internship at DRDO, a prestigious Indian government facility under the Ministry of Defence.
- Chosen as one of the 100 students nationwide for the Prosus AI University Games Hackathon on real-world agentic AI challenges.

CERTIFICATIONS

- **Mathematical Biostatistics Boot Camp 1 & 2** by Johns Hopkins University. [Link](#)
- **Advanced Linear Models for Data Science 1: Least Squares** by Johns Hopkins University. [Link](#)

VOLUNTEER EXPERIENCE

Hamari Pahchan NGO, New Delhi, India

- Matched NGO projects with CSR goals by analyzing partner reports and aligning initiatives for strategic outreach.
- Improved campaign impact by analyzing influencer data and local incidents to refine targeting and engagement

SKILLS

- **Technical Skills:** Python, C++, SQL, NumPy, Power BI, TensorFlow, Excel, Power Point, PyTorch, Matplotlib, Reinforcement Learning (AC, A2C, PPO, SAC), Prompt Engineering, Time-Series, Predictive Modeling, NLP, LLM, LAM, Model Evaluation, Agentic AI, Gen AI, Information Retrieval, API Integration, CNN.
- **Soft & Collaboration Skills:** Analytical Thinking, Communication, Documentation, Version Control (Git), Problem Solving, Adaptability, Creative thinking.